Multimodal Trajectory Prediction with a GRU Decoder

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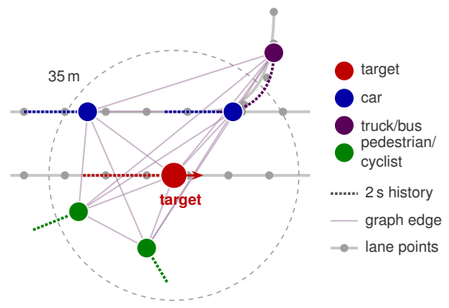
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Motivation and Problem

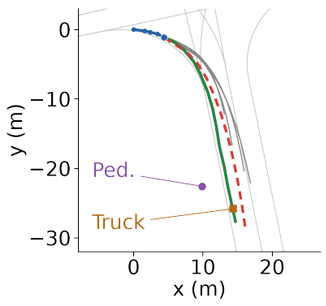
To plan safely, self-driving cars must anticipate where road users go next, and the future is **multimodal**: one past can lead to several futures.

- **Data:** nuScenes [2], target-centred frame
- **Input:** 2 s of all agents within 35 m + lanes within 50 m
- **Output:** K=6 futures of 6 s (12 poses)

Question: what do *interaction*, *map context* and a *recurrent decoder* each add?



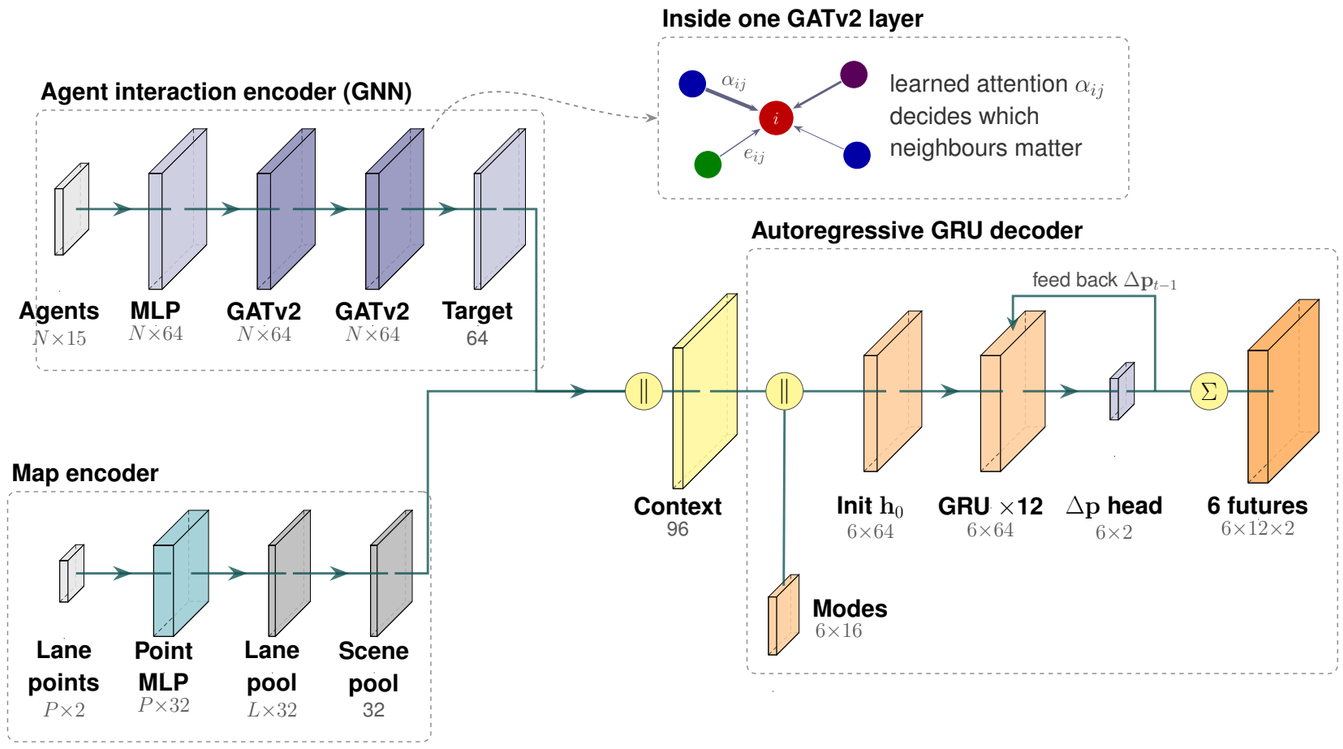
Input: agents as a graph.



Output: truth, best, other modes.

Model: Graph Encoder + Recurrent Decoder

Target-only MLP → **+ GATv2 interactions** → **+ 6 modes** → **+ lane map** → **+ GRU decoder**



Architecture: GATv2 + map encoder feed a GRU decoder (6 modes).

- **Encoders:** agents form a graph processed by GATv2 [1]; lanes are point sets pooled as in VectorNet [4].
- **Decoder:** for each mode k, a GRU predicts the next displacement from the previous one:

$$\mathbf{h}_t^{(k)} = \mathrm{GRU}\big(\Delta\hat{\mathbf{p}}_{t-1}^{(k)}, \mathbf{h}_{t-1}^{(k)}\big)$$

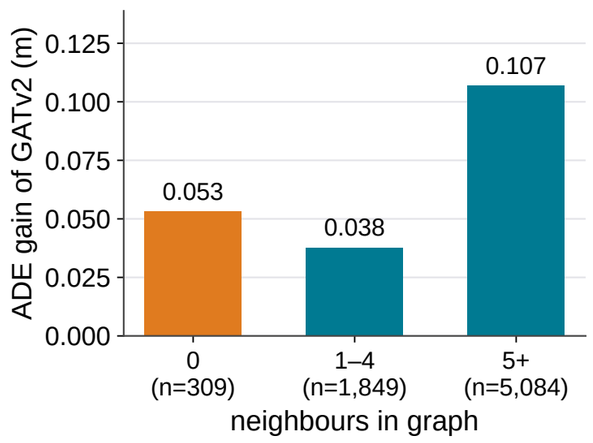
- **Metrics:** error of the *best* of the 6 futures, averaged over time (ADE) or at the end (FDE):

$$\mathrm{minADE}_6 = \min_k \tfrac{1}{12}\textstyle\sum_t \big\|\hat{\mathbf{p}}_t^{(k)} - \mathbf{p}_t\big\|$$

$$\mathrm{minFDE}_6 = \min_k \big\|\hat{\mathbf{p}}_{12}^{(k)} - \mathbf{p}_{12}\big\|$$

- **Loss:** minADE_6 (best-of-K [3]).

Analysis: Does the GNN Use Its Neighbours?



- **Single-mode** GATv2 beats a target-only MLP on average, even with **no neighbours**.
- In the **multimodal model**, removing all neighbours changes minADE_6 by $<$0.001 m (3 seeds).

Figure 1: ADE gain of GATv2 over a target-only MLP, by neighbour count (seed 42).

Analysis: Sharp-Turn Errors

1. Problem: error grows with the turn angle; turns $\geq$30° have **2–3**$\times$ the error of straight driving (grey bars in Fig. 2).

2. What we tried (error before → after; lower is better):

Change	Turns 30–60°	All data	Effect
Add lane map*	3.61 → 3.45	1.72 → 1.54	both better
60 vs 30 epochs*	3.45 → 3.13	1.54 → 1.53	mostly turns
Oversample turns*	3.13 → 2.77	1.53 → **1.56**	overall worse
GRU decoder	**3.09 → 2.73**	**1.53 → 1.47**	**both, all 3 seeds**

Table 1: minADE_6 in m, one-shot MLP decoder unless noted; *seed 2026.

3. Outcome: oversampling hurts overall error; a **GRU decoder improves both**

Results: GRU Decoding Improves Prediction

GRU vs. one-shot MLP decoder (same encoders, 3 seeds):

−3.8% average error (minADE_6)

−6.1% final error (minFDE_6)

−18.1% final error (minFDE_6), 30–60° turns only

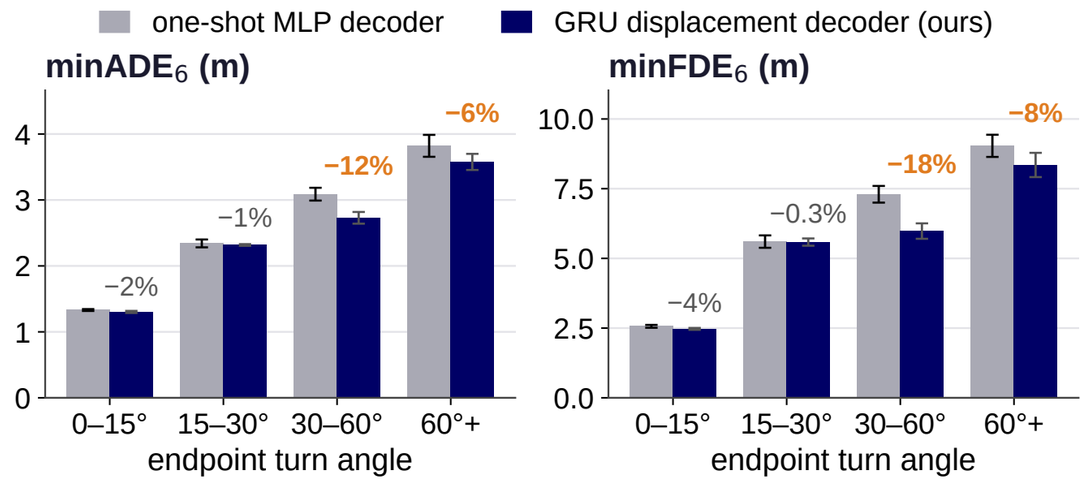
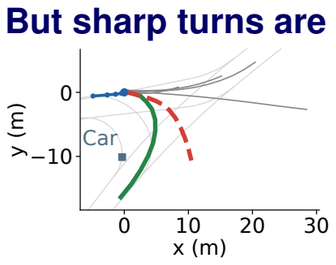


Figure 2: Test error by turn angle (mean ± std). Overall: minADE_6 1.53 → **1.47** m, minFDE_6 3.12 → **2.93** m. The GRU wins in every seed, most in turns.

But sharp turns are still hard.



- A 92° turn from the test set.
- Even the best mode fails: **12.4 m** final error (minADE_6 4.4 m).

Figure 3: GRU failure on a sharp turn.

Conclusion

What does each part add?

- **Map context:** −9.3 % / −13.4 % (ADE / FDE), 3 seeds.
- **GRU decoder:** −3.8 % / −6.1 % (ADE / FDE), most on turns.
- **Interaction:** limited; the multimodal model barely uses neighbours.

Takeaway: lane geometry and step-by-step decoding contributed most.

References

[1] S. Brody, U. Alon, and E. Yahav. How attentive are graph attention networks? In *ICLR*, 2022.

[2] H. Caesar et al. nuScenes: A multimodal dataset for autonomous driving. In *CVPR*, 2020.

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[4] J. Gao et al. VectorNet: Encoding HD maps and agent dynamics from vectorized representation. In *CVPR*, 2020.

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